

Problem 289: Radar Tracking

Difficulty: Medium

Author: Craig Noble, Manassas, Virginia, United States

Originally Published: Code Quest 2026

Problem Background

Radar systems work by transmitting radio waves over long distances, then listening for the “echo” of these waves as they bounce off objects in the distance, such as aircraft. Using this information, a radar system is able to determine the direction and distance to a distant object. Sonar and lidar systems work on similar principles, using sound waves or infrared light rather than radio. By using the information obtained from multiple “pings” against the same object, it’s possible to determine how fast the object is moving, and in which direction.

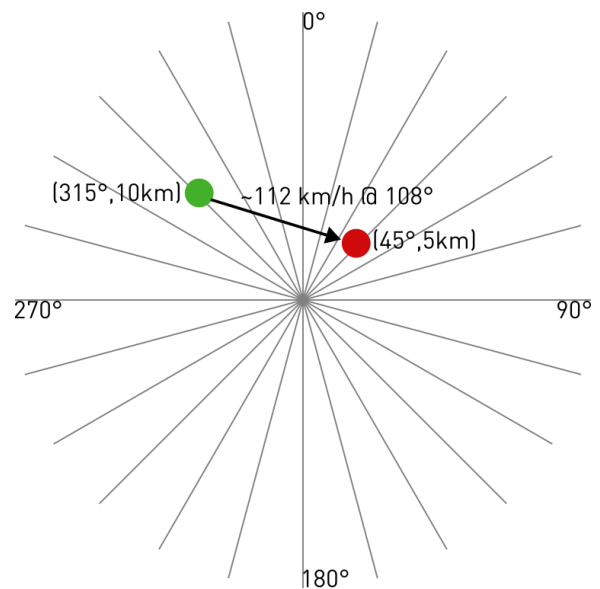
Problem Description

You’re working with a team at Lockheed Martin Rotary and Mission Systems to develop a new radar system for use in civilian air traffic control systems. To make the locations of aircraft easier for air traffic controllers to understand, your team wants to overlay the information obtained from the radar system over top of a satellite image, with each dot indicating a plane showing that plane’s speed and heading. Unfortunately, radar systems work using a polar coordinate system, which can make it difficult to obtain a cartesian heading for a distant, moving object.

You’ll be given two sets of polar coordinates that both pertain to the same aircraft, obtained 6 minutes apart from each other while the aircraft is moving in a straight line at constant speed. Each coordinate will indicate a compass bearing from the air traffic control tower and a distance from that tower in kilometers. Using this data, you’ll need to determine the aircraft’s speed in kilometers per hour, and the heading (direction) of the aircraft.

For example, an airplane was sighted at a bearing of 315° , 10 kilometers away. Six minutes later, it was 5 kilometers away at a bearing of 45° . Using the trigonometry functions provided in the Reference Materials, and the time difference of six minutes, we can determine that the airplane is travelling at about 110 kilometers per hour, at a heading of about 108° .

For this problem, due North is 0° , due East is 90° , and so on; see the image at right for a reference.



Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a single line containing four non-negative integers, separated by spaces, each representing:

- The angle at which radar first detected an aircraft, in degrees between 0 (inclusive) and 360 (exclusive)
- The distance at which radar first detected an aircraft, in kilometers
- The angle at which radar detected the aircraft six minutes after the first sighting, in degrees between 0 (inclusive) and 360 (exclusive)
- The distance at which radar detected the aircraft six minutes after the first sighting, in kilometers

```
3
315 10 45 5
180 20 270 20
255 12 75 10
```

Sample Output

For each test case, your program must print a single line containing the following values, separated a space:

- The speed of the aircraft in kilometers per hour, rounded to the nearest integer
- The heading of the aircraft in degrees, rounded to the nearest integer

Refer to the Reference Materials for details on how to correctly round numbers.

```
112 108
283 315
220 75
```